SAY WATT ROBOTICS · FIELD LOG

Week 2.5 — Cardboard, Then Plastic, Then Brass — The Buoyancy Engine Takes Shape

Sunday, October 18, 2026 · 3:00 – 5:00 PM

Attendance wasn't recorded this week, but the photos tell the story: the buoyancy engine — the part of the float worth 20 of its 70 points — is starting to look like a machine. On the kitchen island, the electrical work moved from paper to a breadboard. An ESP32 is now wired to a TMC2209, a quiet stepper motor driver that turns pulses from the microcontroller into precise steps of a motor. A stepper suits this job because steps are countable: count them and you know exactly how far the plungers moved and how much water the float has taken on. Next to it sat the float's wiring diagram, drawn by hand — a 12-volt battery through a single fuse, a buck converter stepping down to 5 volts for the ESP32, the motor driver and stepper on the 12-volt side with a 470-microfarad capacitor across the supply, and a micro SD card, a pressure sensor, and a light sensor hanging off the ESP32. On the table, the syringe plate showed its whole iteration history in three pieces: a cardboard cutout, a first 3D print, and a second print with a brass lead-screw nut pressed into the center — the nut that will let a turning screw move the plungers together. In another room, the next float part was coming off the printer. The software track worked on the ESP32's data packets, the format the float will use to report what it measured, and the search for a new marketing lead continued.

3	1	470 µF	20
GENERATIONS OF THE SYRINGE PLATE — CARDBOARD, PRINT, PRINT WITH BRASS NUT	FUSE BETWEEN THE BATTERY AND EVERYTHING ELSE	CAPACITOR ACROSS THE MOTOR SUPPLY	OF THE FLOAT'S 70 POINTS RIDING ON THE BUOYANCY ENGINE

WHAT WE LEARNED

- A stepper motor moves in exact steps. Count them and you know how far the plungers traveled, which is how the float knows how much water it is holding without a separate sensor.
- A stepper driver wants a big capacitor across its motor supply. Motors throw voltage spikes back into the supply when they start and stop, and the capacitor soaks them up before they reach the driver.
- A lead screw multiplies force. A small motor turning a screw can push and pull syringe plungers against the water pressure at four meters, and a fine-pitch screw holds its position when the motor stops.
- Draw the power system before you build it. One page showing every voltage, the fuse, and where the 12 V and 5 V sides meet is also the beginning of the System Integration Diagram the judges will ask for.
- Iterate in the cheapest material first. Cardboard tests the layout for free, the first print tests the shape, and the second print adds the hardware.
- Have the pinout open while you wire. The TMC2209's pin diagram was on the laptop the whole time — that is how you keep 12 volts away from a 3.3-volt pin.

DECISIONS MADE

- The buoyancy engine runs on a stepper motor through a TMC2209 driver. Steps are countable, so the plunger position — and the water the float is carrying — is known, not guessed.
- A lead screw drives the syringes. The second printed plate carries a brass lead-screw nut, so turning the screw moves every plunger together.
- The float's power system is on paper: a 12 V battery through a single fuse, a 5 V buck converter for the ESP32, and the motor driver on the 12 V side with a 470 µF capacitor across it.
- Prototype in the cheapest material that answers the question. Cardboard checks the fit; plastic checks the function; the hardware goes in last.

STUDENT LEARNING OUTCOMES

- Wired a TMC2209 stepper motor driver to an ESP32 on a breadboard.
- Drew the profiler's power and signal diagram: battery, fuse, buck converter, ESP32, motor driver, stepper motor, SD card, and pressure and light sensors.
- Iterated the syringe plate from cardboard to two 3D-printed versions, the second carrying a brass lead-screw nut.
- Continued CAD and 3D printing of the buoyancy engine and profiler parts.
- Worked on the ESP32 data packet format for the float's reports.

LOOKING AHEAD

Get the stepper turning the lead screw, then get the lead screw moving syringes. The hand-drawn wiring diagram is the starting point for the float's System Integration Diagram, and the data packets have to carry depth, water temperature, and light with a timestamp — the format MATE scores. The full 2027 competition manual is expected in November; when it lands, every number from the preview mission gets checked against the final rules.

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